Global Renewable Energy in 2025: How Far Have We Come?

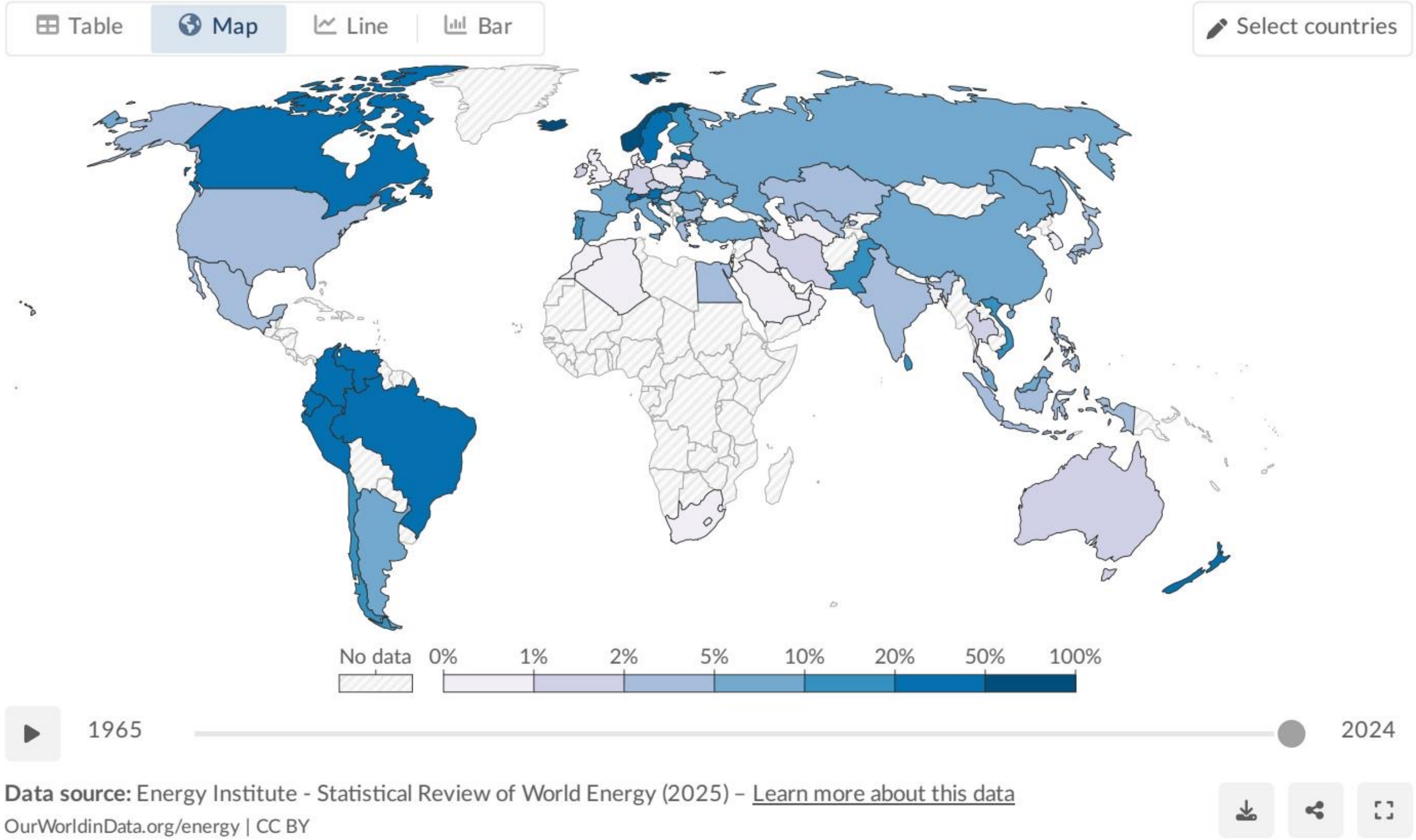
A data-driven overview of global renewable energy trends across primary energy and electricity.

Source: Our World in Data – Renewable Energy

75% of Global GHG Emissions Still Come from Fossil Fuels

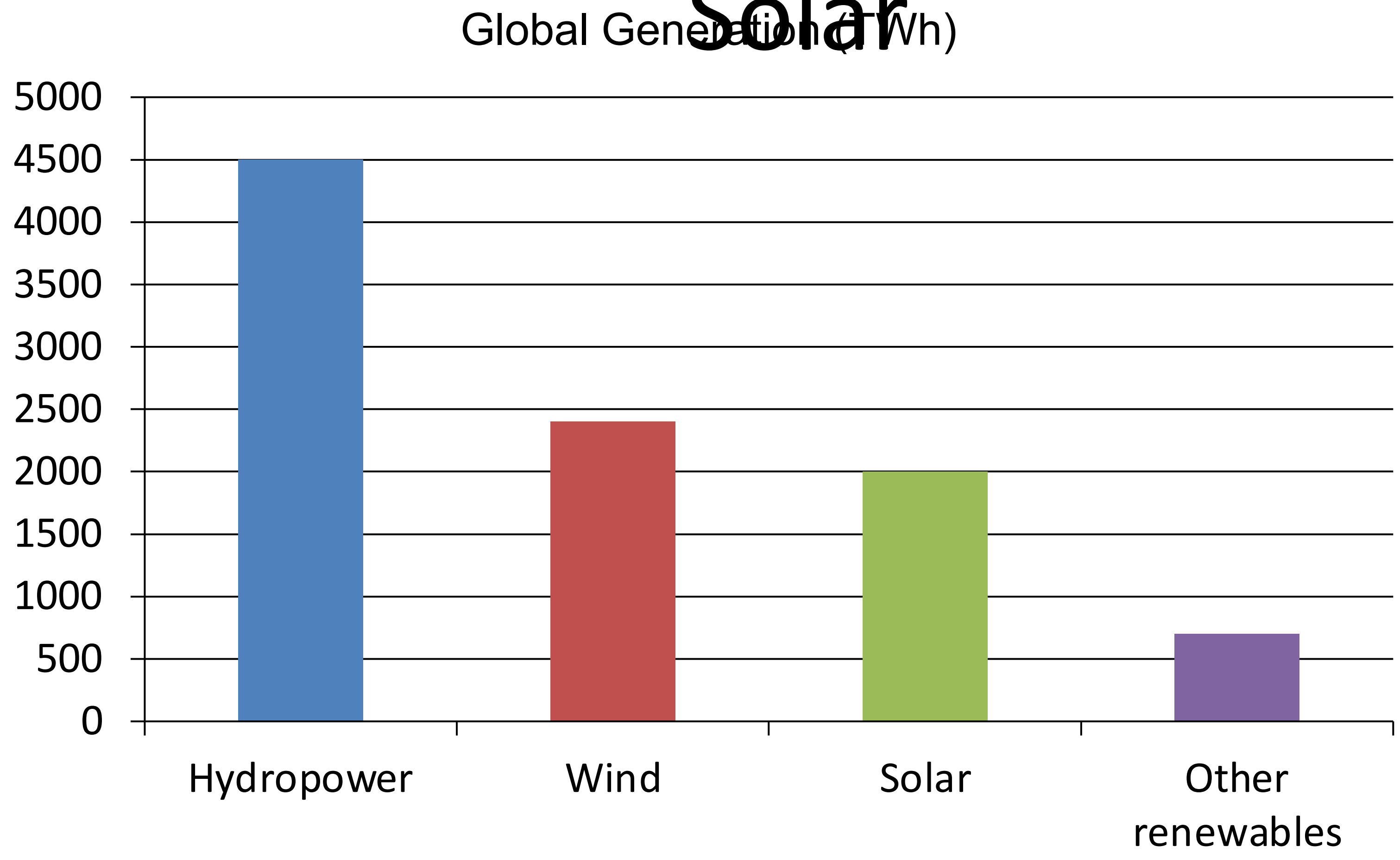
- Fossil fuels have dominated the global energy mix since the Industrial Revolution.
- Three-quarters of global greenhouse gas emissions come from burning fossil fuels for energy.
- Fossil fuel combustion causes at least 5 million premature deaths per year from local air pollution.
- Urgent Action Needed: The world must rapidly shift toward low-carbon energy sources

Renewables Now Supply ~14% of Global Primary Energy



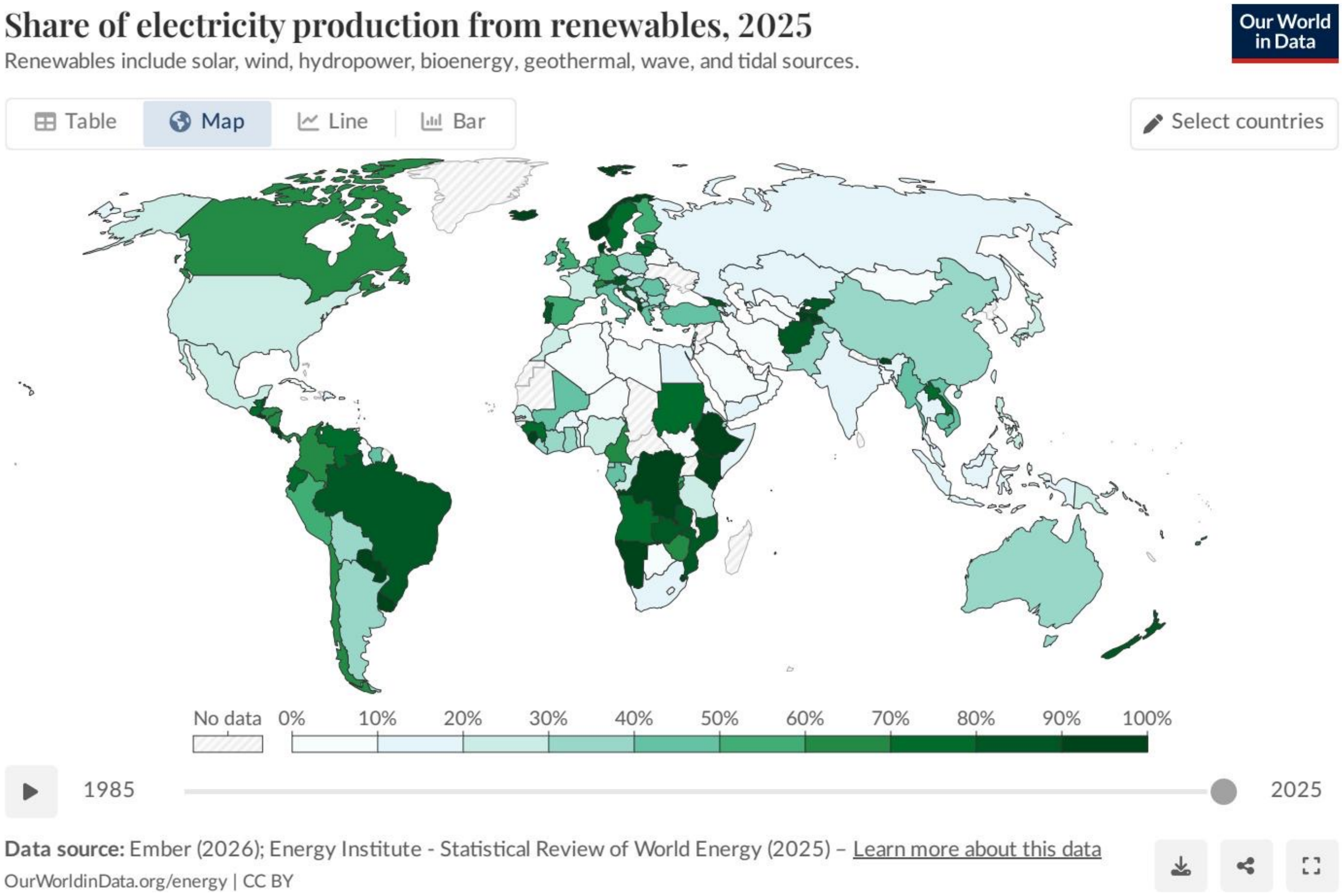
- Approximately one-seventh (~14%) of the world's primary energy comes from renewables (2024, substitution method).
- Primary energy encompasses electricity, transport, and heating combined.
- Because transport and heating are harder to decarbonize, the overall renewable share in the total energy mix is lower than in the electricity mix alone.
- Regional variation is significant: countries like Norway, Brazil, and New Zealand have large shares, while fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



- Total renewable electricity generation is scaling rapidly across the globe.
- Hydropower remains the largest at ~4,500 TWh.
- Wind (~2,400 TWh) and solar (~2,000 TWh) have accelerated dramatically since ~2010.
- Combined, wind and solar now nearly match hydropower's output.
- Other renewables (geothermal, biomass, wave, tidal) contribute ~700 TWh.

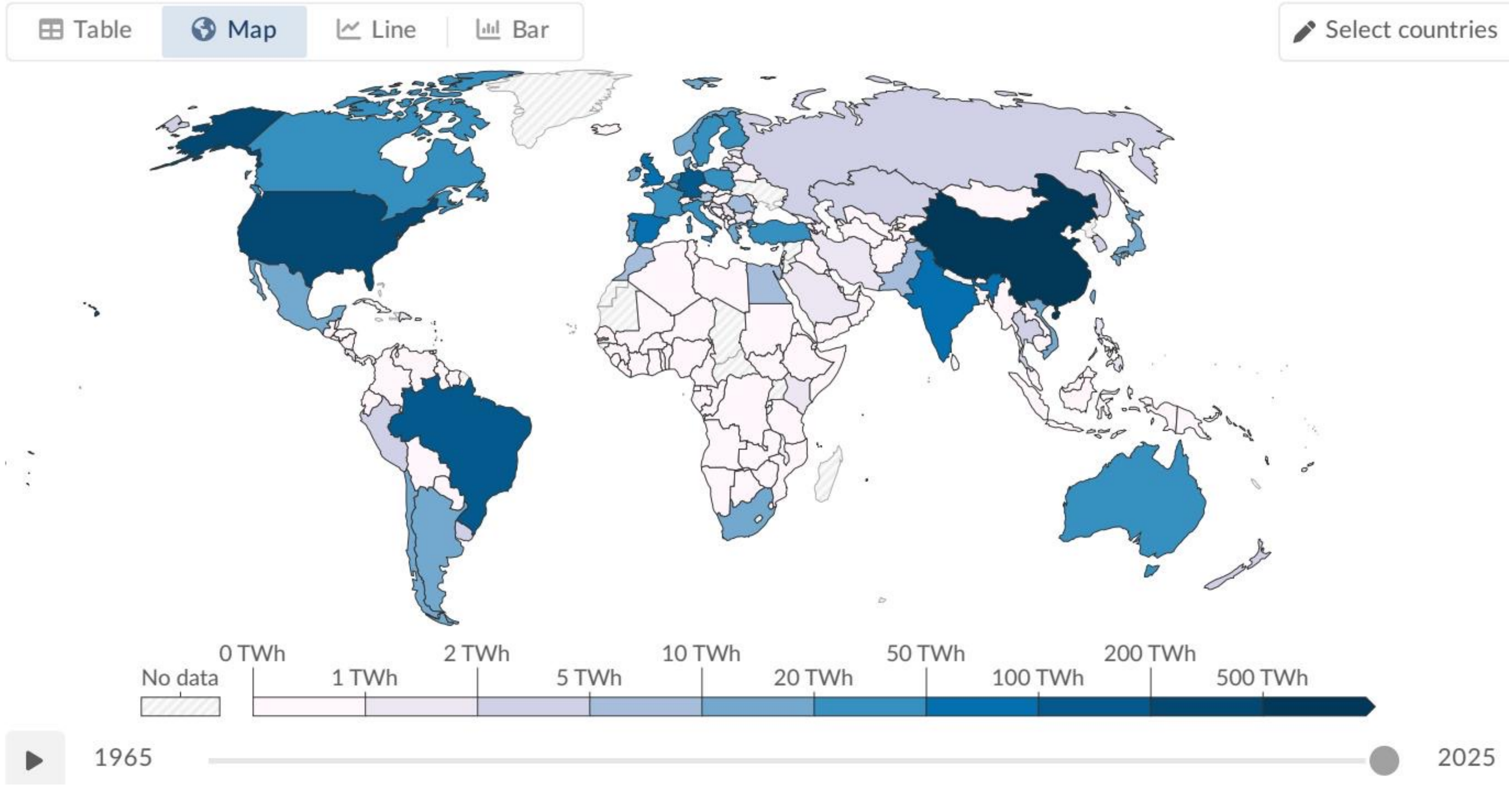
Almost One-Third of Global Electricity Now Comes from



Country/Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

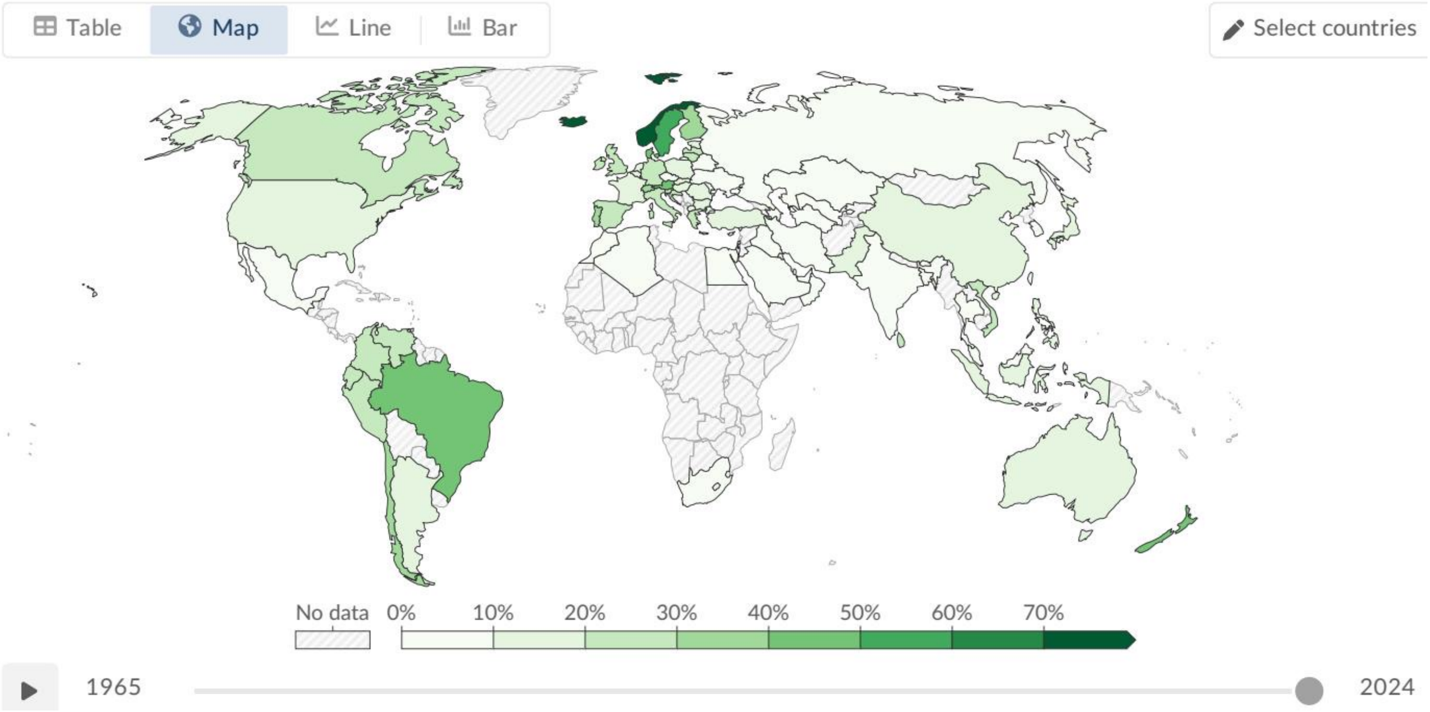
- Globally, almost one-third (~30%) of electricity comes from renewable sources (2025).
- Regional Leaders: Nordics, Brazil, and Canada achieve 65–90%+ shares.
- Laggards: Middle East and parts of Asia remain below 10%, reflecting fossil fuel dependence and limited investment.

Hydropower Remains the Largest Renewable Source at ~50% of



- Hydropower accounts for roughly half (~47%) of all renewable electricity generation globally.
- It has been generating electricity at scale for over a century, providing baseload reliability.
- Top producers include China, Brazil, and Canada.
- However, its future growth potential is limited compared to wind and solar due to geographical and environmental constraints.

Wind and Solar Are the Fastest-Growing Energy Technologies in



Wind Energy:

• ~2,400 TWh generated (~25% of renewable electricity).

• Major contributor in Europe, China, and the United States (includes onshore and offshore).

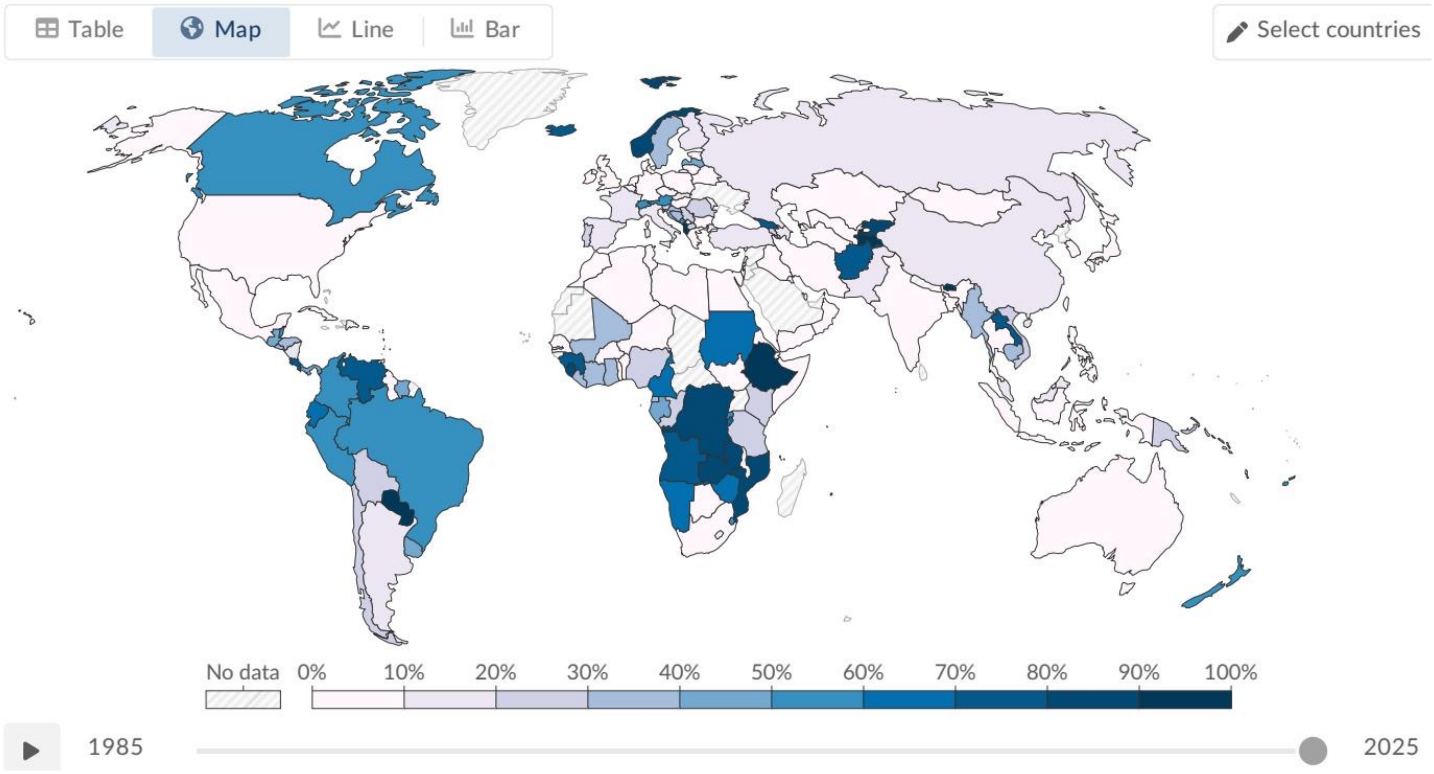
Solar Energy:

• ~2,000 TWh generated (~21% of renewable electricity).

• The fastest-growing source overall, driven by falling costs and policy support.

Together, wind and solar combined now nearly match hydropower output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia &



- The transition to renewables is highly uneven globally.
- Scandinavia and Latin America: Lead the world (65–98% shares) thanks to abundant hydropower resources.
- Europe: Nations like Denmark and Germany have heavily invested to transition via wind and solar power.
- Sub-Saharan Africa: Has minimal renewable electricity infrastructure despite significant potential.
- Middle East: Many nations remain almost entirely dependent on fossil fuels for electricity (e.g., Saudi Arabia <2%).

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- Significant Milestones Reached: ~14% of global primary energy and ~30% of global electricity now come from renewable sources.
- Hydropower is still the dominant source (~50% of renewable electricity), but wind and solar are the fastest-growing energy technologies in history and are rapidly catching up.
- Stark regional disparities persist, with Scandinavia and Latin America leading (60–